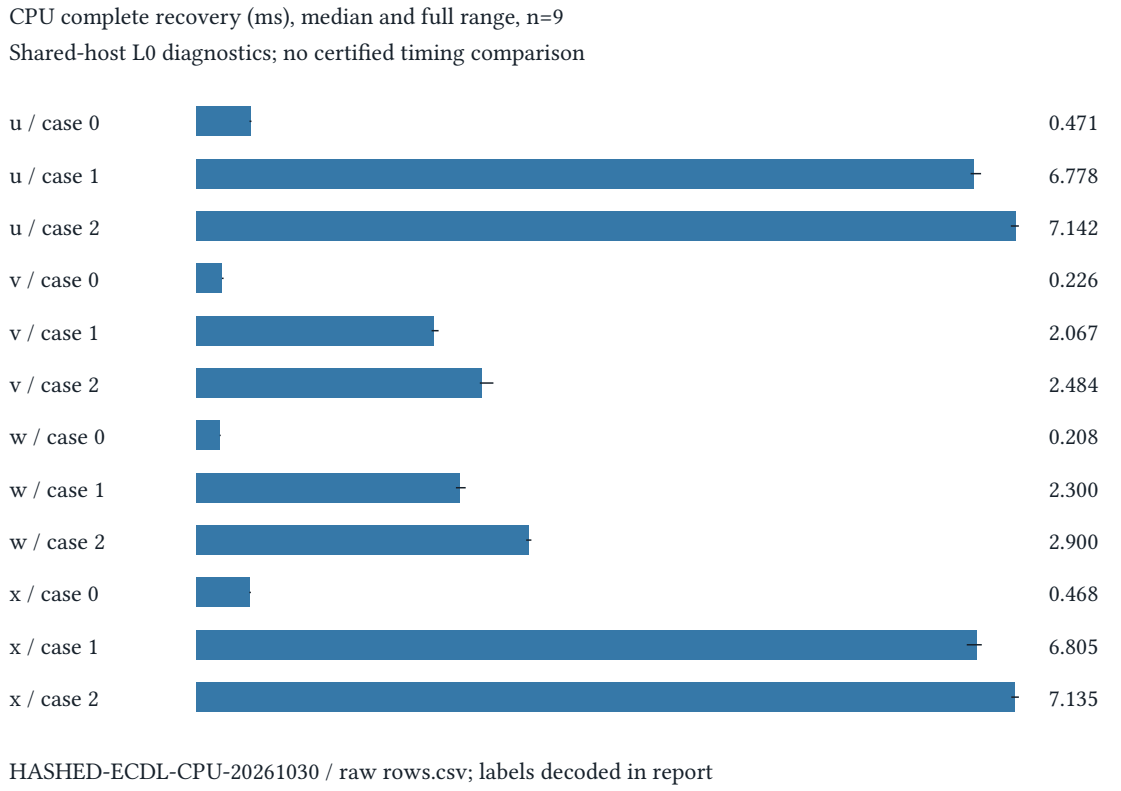
CPU order and residual arithmetic screen

Observations only; no certified latency claim or hypothesis promotion. Run: `hashed-ecdl-cpu-round4-20261030`, October 29, 2026 PDT.

Baseline u is the optimized CPU implementation at `2bd4588fba60`; candidate v is `19748cd3248d`, including exact order-witness construction, Euclidean inverses, fixed-base scalar tables and conditional residual subtraction. Candidate w uses batch 1. This is a bundled comparison with unchanged inputs and output semantics. Host: Intel Xeon Platinum 8488C, one software thread, 2 vCPUs / 4 GiB reserved; shared EKS VM, L0. Compiler and release settings match across arms.

Fixture	Baseline u, ms	Candidate v, ms	A/A x, ms
8-bit / 3 slots	0.471432	0.225746	0.467688
16-bit / 3 slots	6.777632	2.067202	6.804898
16-bit / 4 slots	7.142432	2.484374	7.134884

Recorded medians from 9 repetitions after 1 warmup. Derived u/v median ratios: 2.088, 3.279, 2.875. These do not meet the further 10x target and are not strict-isolation performance evidence. Do not pool durations with other hosts.



All 120 executions completed: 108 measured, 12 warmups. Complete certificate bytes agree across arms. Every execution checks its known scalar, replays the certificate and rejects a corrupted scalar. The frozen native analyzer and file-verifying export check pass. No failed or missing cells were discarded.

Complete intervals include constructor and known-target setup, scalar/MITM tables, all attempted trials, relation collection, elimination, both certificate checks and serialization. Process launch, archive I/O and negative control are outside the clock. Table capacity counters are not total peak RSS.

On the 16-bit candidate fixtures, constructor medians are 0.008617/0.008467 ms, but base enumeration is 0.220313/0.244822 ms and separate replay is 0.333475/0.385913 ms. Oracle medians are 0.586966/0.823394 ms. On-demand square roots replace the exhaustive lookup table; further changes need fresh measurement. Largest feasible CPU scale, matched rho/external comparisons, strict isolation and independent review remain unresolved. The 20–40-bit ladder remains separate.

Protocol: `0e878acaf6d00b2ca54b3db91e2921fd9d93f69cd8342ef4b5cdedec5d0843e2`. Raw rows, summaries, certificates and checked export: `round4/results/`. Build identities: `round4/results/build.json`; terminal receipt: `round4/status.json`.